Conservation of Energy

Question 1 (Conceptual): A roller coaster car of mass 500 kg starts from rest at the top of a 50 m high track. Neglecting friction, compare its mechanical energy at the top and at a point 30 m below the start. Explain using the work–energy theorem.

Question 2 (Calculation): A 2.0 kg block is released from rest at the top of a frictionless incline of height 10 m and length 20 m.

(a) Identify the equation for conservation of mechanical energy.
(b) Calculate the block’s speed at the bottom.
(c) If a constant friction force of 15 N acts along the incline, how much mechanical energy is lost to heat?
(d) What is the block’s speed at the bottom with friction?